Final Group Project

Introduction to Data Analysis with Python

Cecilia GRAIFF

Cross-Sectional Study of GDP and Suicide Rates Across 160 Countries

Group 4

Viola POMILI POMILI

Nethra JAYGOPAL

Neda MILTINYTE

1. **Introduction and motivation**

Suicide is considered a major global health issue, claiming thousands of lives worldwide each year. Economic conditions are often considered a potential determinant of how high suicide rates vary across countries, although studies show varying results. While some demonstrate no correlation between GDP and suicide rates,[1] others see trends of correlation, such as lower GDP per capita correlating with higher suicide rates and vice versa. This shows that almost 80% of global deaths by suicide occur in low and middle-income countries.[2] Researchers also often distinguish the GDP and suicide rates by different time periods, age or gender. In fact, recent studies have shown that between the years 2018 and 2023, suicide rates for female youth increased faster than those of male youth."[3] Conversely, in 2008, the increase in suicide for men after 2008 was much larger than it was for females, concluding that this might be linked to the 2008 economic crisis."[4] Therefore, economic conditions might have an impact on suicide rates in different countries, especially during recessions or other economy-wide shocks.

There are also interesting findings when it comes to growth, with some studies concluding that "Economic growth has a protective effect against suicide."[5] However, some studies show that if economic growth does not lead to overall prosperity[6] and if growth is not accompanied by adequate infrastructures for mental health services, suicide rates might trend up.[7] Therefore, it is not clear whether high GDP per capita can be associated with low suicide rates. GDP per capita serves as a good variable for data collection because it is linked with economic well-being and living standards (unemployment, poverty) and is a standard measure across countries. Identifying whether there is a correlation between the GDP per capita and suicide rates is important to assess what kind of public health measures should be applied. For instance, in developing countries, public health interventions might be more suitable, but in high-income countries, preventive measures based on the medical model might prove more useful.[8]

This project analyses the relationship between economic development and suicide rates across countries using a dataset obtained from Kaggle titled "GDP per Capita and Suicide Rates." The dataset compares countries' gross domestic product (GDP) per capita with their crude suicide rates to explore whether economic conditions are associated with different suicide rates to explore whether economic conditions are associated with differences in suicide rates.

[1] Cao, 'Research on the Relationship between Suicide Rates and GDP: A Statistical Analysis'.
[2] United Nations, 'World Mental Health Day'
[3] Carretta et al., 'Gender Differences in Risks of Suicide and Suicidal Behaviors in the USA'.
[4] Ibid.
[5] Lee et al., 'Economic Growth and Suicide Rates'.
[6] Ibid.
[7] Blasco-Fontecilla et al., 'Worldwide Impact of Economic Cycles on Suicide Trends over 3 Decades'.
[8] Ibid.

The dataset contains observations for 160 countries for the years 2000, 2005, 2010, 2015 and 2016. The suicide rate is measured as the number of suicide deaths per 100,000 inhabitants, calculated by dividing the number of suicide deaths in a year by the population and multiplying the result by 100,000. Although the dataset is hosted on Kaggle, the data originates from international sources. The suicide rate data comes from the Global Health Observatory of the World Health Organisation (WHO), while GDP per capita data is sourced from the World Bank Data Bank.

For this analysis, we decided to focus specifically on the year 2015. Limiting the dataset to a single year simplifies the data cleaning and preprocessing process, an important step in the data analysis pipeline. This approach follows the data analysis pipeline presented in the course, where data must be collected, cleaned and inspected before performing exploratory data analysis and visualisation. Focusing on a single year also allows for clearer cross-country comparisons and simplifies the interpretation of the relationship between GDP per capita and suicide rates. We believe that even though we are analysing the data for 2015, our findings can be relevant for the year 2026. This is because both time frames are similarly 6 years after large economic shocks: the 2008 financial crisis and the 2020 COVID pandemic.

Our choice not to include gender or age serves as one of the main limitations of our study. Even though we understand that controlling for gender and age might reveal differences in the correlation between GDP and suicide rates, we decided not to include an additional layer of variability. We wish to examine the crude rates of suicide, while controlling just for GDP per capita, since adding other variables could introduce error or bias.

2. Research question and Methodology

The central research question guiding this project is: ***What is the relationship between GDP per capita and suicide rates across countries in 2015?*** More specifically, the analysis investigates whether countries with higher levels of economic development tend to experience higher or lower suicide rates.

Two hypotheses were formulated to frame the investigation:

- **H_0 (Null Hypothesis):** There is no relationship between GDP per capita and suicide rates across countries in 2015.
- **H_1 (Alternative Hypothesis):** There is a negative relationship between GDP per capita and suicide rates, meaning that countries with higher GDP per capita tend to have lower suicide rates.

The unity of analysis is the country. Each observation represents a single country in 2015, making this a cross-sectional study that enables direct comparison across nations at the same point in time. The dependent variable is the crude suicide rate per 100,000 inhabitants. The independent variable is GDP per capita in 2015 US dollars, used as a proxy for economic development and average living standards.

3. Justification of the Analytical and Visualisation Method

The analytical and visualisation methods used in this project follow the data analysis pipeline presented in the course, moving from data cleaning and summary statistics, through graphical and multivariate exploratory data analysis.

Our first step was to clean the data. We first checked for missing values and duplicate values in the rows to eliminate any errors in our final calculations. Once we did, we could move on to our next step of outlier handling. This was applied to the suicide rate before visualisation. The IQR, the range of the middle 50% of the data, is robust against extreme values, since the top and bottom quarters of the data can be moved without affecting it. The IQR ensured that the patterns observed in our visualisations and statistics genuinely reflect cross-country trends, rather than being distorted by a small number of anomalous cases.

We then decided to band our data by GDP. We divided countries into three categories: Low (below $4,000), Medium ($4,000–$8,000), and High (above $8,000) GDP per capita. These thresholds were chosen to reflect meaningful and commonly used distinctions in economic development: low-income countries with limited public services and infrastructure, middle-income countries in transition, and high-income countries with more developed welfare systems. We decided to band them in these categories as they were the most even applications to our data. By converting the continuous GDP variable into bands, we were able to apply these methods and directly compare how suicide rates differ across income levels.

Our next step involved creating summary statistics. We did this as our first analytical step to understand the structure and the distribution of data. For each variable, we computed measures of central tendency (mean, median and mode) as well as measures of spread, including standard deviation, variance and interquartile range. We also computed the correlation and covariance between GDP per capita and the suicide rate in 2015. Covariance indicates the direction of the relationship between two variables, while correlation standardises this measure to a range of -1 to +1, making it easier to interpret the strength of the linear relationship regardless of the units of the variable. We then checked whether our outlier handling worked, which we confirmed it did.

We then visualised our results through various graphs. A boxplot was used to compare the distribution of suicide rates across GDP bands. It displays the interquartile range, the median, and whiskers extending to the last values within 1.5 times the IQR, with any points beyond that treated as outliers. Boxplots are particularly effective for comparing distributions across categories, across the three income bands, and for identifying skewness within each group.

We created scatter plots to visualise correlations, patterns and outliers between two numerical variables, where each point represents a single observation's values on the x-axis and y-axis. Since our research question is fundamentally about the relationship between two continuous variables (GDP per capita and suicide rate), scatter plots are essential. We produced three variations. A basic scatter plot displays the overall relationship across all countries. A second scatter plot adds a linear trend line as a tool to detect directional trends between the two variables. A third grouped scatter plot colours countries by GDP band, allowing sub-group patterns to become visible within the same figure. Finally, three side-by-side scatter plots, one per GDP band, as a technique for comparing the same type of plot across multiple subsets of the data.

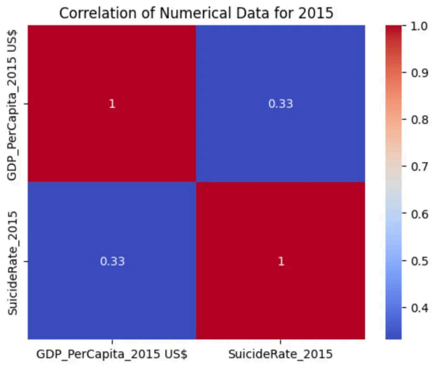
A histogram was used to examine the distribution of the dependent variable, the suicide rate. Histograms are suited to continuous variables: they divide the values into bins and plot those bins, allowing us to check the shape of the data's distribution, whether it is symmetric, right-skewed, or left-skewed.

A bar chart of the average suicide rate per GDP band was used to compare mean values across the three income categories. Bar plots are well-suited for comparing discrete values across categories, placing categories on one axis and values on the other. Bar charts communicate headline findings simply and are accessible.

A correlation heatmap was generated to display pairwise correlations among all numerical variables in the 2015 dataset. Heatmaps plot correlation coefficients in a matrix format, where the colour scale allows the reader to immediately spot strong positive (close to 1) or negative (close to -1) relationships and identify variables with no linear relationship (close to 0). This makes the heatmap a powerful tool for non-graphical analysis presented in graphical form, providing a broader view of the data structure beyond the single GDP-suicide rate pair at the centre of our research question.

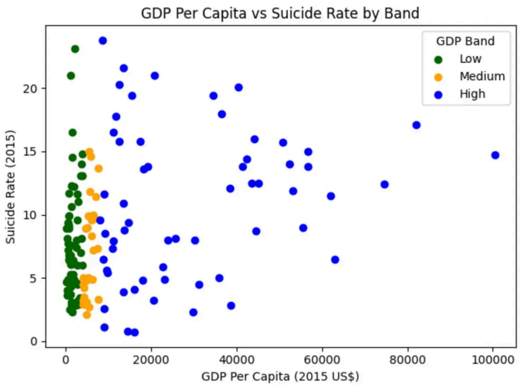
4. Analysis of our research

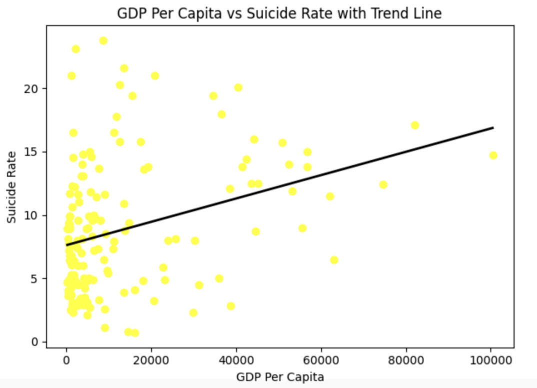
Through our research, we hoped to investigate if a negative relationship between GDP per capita and suicide rates. This would entail that higher GDP countries (in our study, those countries with per capita GDP higher than USD 8,000) have a lower suicide rate and vice versa. We found that the correlation coefficient was low, meaning there is no strong linear relationship between GDP per capita and suicide rates. Our correlation coefficient or 'R' was 0.2440908477485541, indicating a weak positive relationship. The correlation coefficient describes the direction and strength of the relationship. Thus, for every increase in GDP, suicide rates very slightly increased. However, the relationship is too weak in our data to be meaningful. R^2 describes how much of the outcome (suicide rates) is explained by the variable (GDP per capita). The R^2 is 0.06 or 6%. Hence, GDP increases explain only 6% of the variation in suicide rates. The other 94% could be explained by other factors such as culture, mental health services, conditions of work and education, and region. More research needs to be done in order to definitely determine this. This supports our H0 that there is no significant relationship between the two.



We hypothesised that this might have been due to the extreme outliers in our dataset (countries with extremely high GDP per capita and high suicide rates) and conducted a test to see if this would change our results as well. We did this through the Inter Quartile Range measurement. We chose this method because it is based on cutoffs on the spread of data, making it the most statistically reliable approach. We calculate the middle 50% of the data and set boundaries at 1.5 times the IQR above and below, to ensure that only extreme values were removed, leaving the rest of the dataset intact for analysis. A clear example of an outlier is Switzerland, with a GDP per capita of USD 82,081yet recorded a suicide rate of 17.1 per 100,000 people. This was unusually high, and hence we decided to remove this. As the correlation map shows, after outlier removal, we see a correlation coefficient of 0.33, which is still too weak to support the H1 hypothesis, evidenced by the blue on the colour map.

Our scatter plot shows similar results. The widespread high-income countries (in blue) explain why the correlation is weak. Wealthier countries behave unpredictably, with some having low suicide rates and others having high ones. The plots range from close to 0 all the way to as high as 24 per 100,000 inhabitants. The clusters of low and middle-income countries (green and orange) show that these are more consistent and predictable. However, clearly through the scatter plot, there is no clear downward trend from left to right, which directly contradicts our H1 hypothesis that there is a negative correlation between high-income countries and suicide rates. We also map a scatter plot with a trend line to clearly demonstrate the weak positive relationship.





Our summary statistics also reveal rich data. The mean suicide rate across all the countries was 9.4 per 100,000. However, the median is noticeably lower, which shows that the small number of countries with higher-than-normal suicide rates was pulling the average upward. The standard deviation of 6.27 also confirms that suicide rates vary considerably across countries, showing that there is no single rate globally. The 25th and 75th percentiles show that most countries fall within a narrow band of suicide rates between 4.8 and 12.5. This confirms that extremely high rates are rare and concentrated in a small number of outlier countries.

Hence, we would like to conclude by showing that our H1 hypothesis was not supported. Thus, we conclude by confirming no significant relationship between GDP per capita and suicide rates among the 160 countries surveyed in 2015. Our visualisations reinforce this conclusion, demonstrating wide variations and concentrated outliers. Our results also align with several studies cited in the introduction, which found mixed or no correlation between GDP and suicide rates. GDP per capita remains just one of the factors studied, and future research must cover variables such as cultural factors, access to mental health services, and social support.

BIBLIOGRAPHY

Blasco-Fontecilla, Hilario, M. Mercedes Perez-Rodriguez, Rebeca Garcia-Nieto, et al. ‘Worldwide Impact of Economic Cycles on Suicide Trends over 3 Decades: Differences According to Level of Development. A Mixed Effect Model Study’. *BMJ Open* 2, no. 3 (2012): e000785. https://doi.org/10.1136/bmjopen-2011-000785.

Cao, Yifa. ‘Research on the Relationship between Suicide Rates and GDP: A Statistical Analysi’. *Highlights in Science, Engineering and Technology* 88 (2024).

Carretta, Rachel F., Sherry A. McKee, and Taeho Greg Rhee. ‘Gender Differences in Risks of Suicide and Suicidal Behaviors in the USA: A Narrative Review’. *Current Psychiatry Reports* 25, no. 12 (2023): 809–24. https://doi.org/10.1007/s11920-023-01473-1.

Lee, Dong-Wook, Yun-Chul Hong, Je-Yeon Yun, Soo-Hyun Nam, and Nami Lee. ‘Economic Growth and Suicide Rates: Differential Accumulated Effects’. *PLOS One* 20, no. 7 (2025): e0327630. https://doi.org/10.1371/journal.pone.0327630.

United Nations. ‘World Mental Health Day 10 October 2019’. 2019. https://www.ohchr.org/en/statements-and-speeches/2019/10/world-mental-health-day-10-october-2019#:~:text=According%20to%20the%20latest%20statistics,a%20large%20percentage%20of%20suicides.